METHOD STATEMENT

# Addmix HPS — Hydroxypropyl Starch Ether Additive for Dry-Mix Mortars

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|---|---|---|---|
| **Project Name** | | **Document No.** | OMX-MS-ADDMIXHPS |
| **Main Contractor** | | **Revision** | 01 |
| **Consultant** | | **Date of Issue** | Nov 2026 |
| **Product** | Addmix HPS | **Status** | For Approval |

## 1. Purpose

This method statement defines the procedure, resources, quality checks and safety measures for the receipt, storage, dosing and quality control of Addmix HPS in the production of dry-mix mortars.

## 2. Scope of Work

This method statement covers the receipt, storage, handling, dosing in dry blending and quality control of Addmix HPS.

- Receipt and inspection of the material.
- Storage and handling in the plant.
- Laboratory trials to set the dosage.
- Dosing and dry blending in production.
- Quality control of finished products.

## 3. Definitions

| Term | Meaning |
|---|---|
| **Dry mix** | Factory-blended mortar powder to which water is added on site. |
| **Dosage** | Additive quantity as a percentage of the total dry-mix weight. |
| **Water retention** | Ability of the fresh mortar to keep its mixing water. |
| **Open time** | Time during which tiles or layers can still be placed. |

## 4. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| **Production Manager** | Overall responsibility for planning and compliance with approved formulations. |
| **Laboratory / R&D** | Sets and approves the dosage by trials. |
| **Production Supervisor** | Supervises weighing, blending and batch records. |
| **QC Inspector** | Tests incoming material and finished products. |
| **HSE Officer** | Checks PPE, dust control and safe handling. |

| Role | Responsibility |
|---|---|
| **Storekeeper** | Receives and stores bags under the stated conditions; controls batch numbers. |

## 5. Materials

| Material | Description |
|---|---|
| **Addmix HPS** | 20 kg and 25 kg bags; 20 ft container 12 – 14 t, 40 ft container 24 – 28 t. |

### 5.1 Key product data

| Property | Typical Value |
|---|---|
| **Type** | Hydroxypropyl starch ether |
| **Appearance** | White powder |
| **Bulk density** | 500 – 600 g/L |
| **Moisture** | ≤ 8 % |
| **Ash content** | 12 ± 2 % |
| **pH** | 8 – 10 |
| **Viscosity (5 % solution)** | 2,000 – 3,000 mPa·s |
| **Shelf life** | 12 months |

## 6. Tools & Equipment

- Calibrated scale
- Ribbon or plough-share mixer
- Sieve
- Laboratory mixer
- Flow table
- Test equipment
- Dust extraction at the dosing point
- Sample containers for retained samples

### 6.1 Personal protective equipment

- Dust mask
- Goggles
- Gloves
- Overalls

## 7. Storage & Handling

- 12 months in original, unopened bags. Store dry, protected from moisture and heat.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

## 8. Pre-Start Requirements

- Formulation with approved dosage available.
- Mixer clean and dry.
- Scale calibrated.
- Dust extraction working.
- Team briefed on dosing and dust safety.

## 9. Laboratory Trial

1. Prepare laboratory batches at three dosage levels within the recommended range.
2. Test water demand, workability, water retention, open time, sag/slip and adhesion as relevant.
3. Select the dosage that meets the product specification.
4. Record the approved dosage in the formulation.

## 10. Preparation of Material

1. Check the batch number and expiry date and record them.
2. Keep bags closed until use.
3. Weigh the additive on a calibrated scale.

## 11. Dosing & Dry Blending

1. Charge cement, sand and fillers into the mixer.
2. Add the weighed additive with the other minor components.
3. Blend dry until fully homogeneous (typically 3 – 5 minutes, as per plant procedure).
4. Discharge and pack the dry mix.
5. Record quantities in the batch record.

## 12. Batch Release

- Test each batch for the key properties.
- Release only when results meet the specification.

- Keep a retained sample of each batch.

## 13. Quality Control & Inspection

| Check | Stage | Acceptance Criteria | By |
|---|---|---|---|
| **Material** | On receipt | White powder, dry, within shelf life | QC Lab |
| **Dosage** | Each batch | As per formulation | Production Supervisor |
| **Product** | Each batch | Within specification | QC Lab |

### 13.1 Inspection and test plan

| Activity | Inspection / Test | Frequency | Record |
|---|---|---|---|
| **Material receipt** | Appearance, moisture, batch | Each delivery | Goods receipt report |
| **Weighing** | Dosage | Each batch | Batch record |
| **Finished product** | Workability, water retention, adhesion | Each batch | QC report |

## 14. Defects & Remedies

| Defect | Likely Cause | Remedy |
|---|---|---|
| **Lumps in mortar** | Additive added to water; poor blending | Add to dry mix; extend blending |
| **Poor water retention** | Under-dosing | Adjust dosage within range |
| **Sticky / heavy mortar** | Over-dosing | Reduce dosage |

## 15. Health, Safety & Environment

| Hazard | Control Measures |
|---|---|
| **Combustible dust** | Avoid dust clouds; no ignition sources; dust extraction |
| **Dust inhalation** | Dust masks; extraction |
| **Slippery floors when wet** | Clean spills dry first |
| **Manual handling of bags** | Correct lifting technique |

- Combustible dust: avoid dust clouds and ignition sources.
- Wear a dust mask, goggles and gloves when handling.
- Keep the work area clean of dust deposits.
- Eyes: rinse with clean water and seek medical advice if irritation persists.
- Skin: wash with soap and water.

### 15.1 Waste & environment

- Collect empty containers and residues in designated bins.

- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

## 16. Cleaning

- Clean equipment with dry methods first, then water.

## 17. Limitations

- Do not dissolve directly in water.
- Store dry; protect from moisture.
- Any change to the dosage must be approved by the laboratory.

## Appendix A — Inspection Checklist

| Inspection Item | Result | Remarks |
|---|---|---|
| **Material received and inspected** | ☐ Yes ☐ No | |
| **Dosage approved** | ☐ Yes ☐ No | |
| **Additive weighed** | ☐ Yes ☐ No | |
| **Dry blended** | ☐ Yes ☐ No | |
| **Batch tested** | ☐ Yes ☐ No | |
| **Batch record completed** | ☐ Yes ☐ No | |
| **Retained sample kept** | ☐ Yes ☐ No | |

| Inspected by | Signature | Date |
|---|---|---|
| | | |